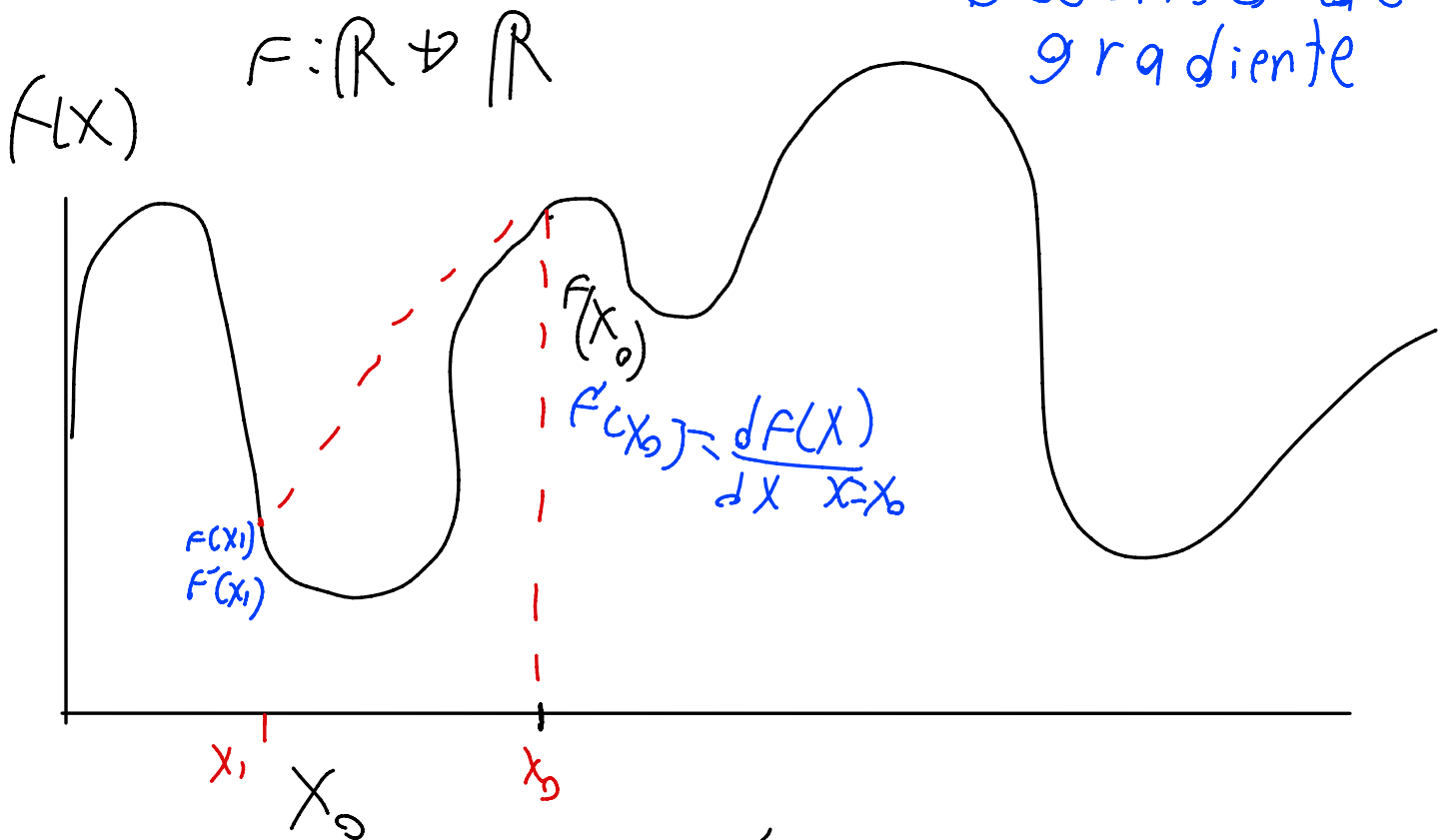


Desenho de gradiente



$$\begin{aligned} x_1 &\leftarrow x_0 - \eta F'(x_0) \\ x_2 &\leftarrow x_1 - \eta F'(x_1) \end{aligned} \Rightarrow x_{k+1} \leftarrow x_k - \eta F'(x_k)$$

$$\theta_{k+1} \leftarrow \theta_k - \eta \underbrace{\nabla_{\theta} F(\theta) \Big|_{\theta=\theta_k}}_{\nabla F(\theta_k)}$$

$$b_{k+1} \leftarrow b_k - \eta \frac{\partial}{\partial b_k} \text{E}_{in}(w_k, b_k)$$

$$w_{j,k+1} \leftarrow w_{j,k} - \eta \frac{\partial}{\partial w_j} \text{E}_{in}(w_k, b_k)$$